

Manual

Power Data Distribution EMG-PDD 8.3

Version 1.0.23049

Last changed on: 01.09.2020

Copyright

©Copyright 2020 All rights reserved.

SAP® and R/3® are registered trademarks of SAP AG.

WINDOWS® is a registered trademark of Microsoft Corporation.

MPDV® and HYDRA® are registered trademarks of MPDV Mikrolab GmbH.

ORACLE® is a registered trademark of ORACLE Corporation, California, USA.

Copying and distribution of this documentation or any part thereof, for any purpose or in any form, is prohibited without prior written permission from MPDV Mikrolab GmbH.

The information contained in this documentation is subject to change without prior notice.

Contents

1	Overview: Real-Time Process Data Monitoring	4
2	Configuration PDV Visualization	5

1 Overview: Real-Time Process Data Monitoring

Overview

Purpose

The product "PDV-RPM 8.3" includes functions for the online visualization of data on the MOC client. You can visualize the process data that is published via MQTT broker. The logical channels and the collection requests that are actively performed at the machines specify the data that can be displayed.

To visualize data, predefined display instruments and a trend graph are available. You can integrate special layouts into the system. Via the relevant editors, you use these layouts to visualize machines or equipment.

Integration

For the product PDV-RPM 8.3, the following requirements must be fulfilled:

- machine interface for data collection in the Process Communication Controller (PCC),
- data must be published via the MQTT broker,
- the basic PDV-PDM package including collection rules must be available.

Features

This product provides the following functions:

- Permanent display of process data of one or more machines and systems. You can define your own layouts or have them generated via customization.
- Display of real time measured values of a machine using predefined gauge charts. You can select the displayed characteristics in an interactive manner.
- Display of real time measured values of a machine as trend line. The last 50 measured values of a process characteristic are visualized online as continuous trend graph.

2 Configuration PDV Visualization

Purpose

This document describes how to configure layouts for the PDV online visualization on the terminal (AIP 8.2) and the MES Operation Center (MOC).

Requirements

The online visualization via MQTT requires an active license PDV-RPM, AIP-PDV. For the terminal or the MOC, the following requirements must be fulfilled:

MOC:

- Configure the storage path in the HYDRA paths with a key "pdvlay" (Go to: System administration → System settings → Paths).

Terminal:

- The terminal version 8.2.1.1 is required.

Procedure

General files

MPDV provides 5 default layouts as of Hydra service pack 12. The layouts include the following configurations:

- Mixed layout with all available controls
- Layout with line charts (*Charts*)
- Layout with digital displays (*Digital*)
- Layout with level indicator (*Linear*)
- Layout with gauge chart (*Gauge*)

The supply also includes a configuration file "MachineLayoutConfiguration.xml". This file lists the available layouts for the different machines.

Configuration of the layout assignment ("MachineLayoutConfiguration.xml")

Sample configuration:

```

<Allocations>
  <LayoutsForAllMachines>
    <Layout id="Global" filename="DefaultLayout.xml">Mix Standard Layout</Layout>
    <Layout id="ALL_1" filename="default_4x4_gauge.xml">Gauge 4x4</Layout>
    <Layout id="ALL_2" filename="default_4x4_digital.xml">Digital 4x4</Layout>
    <Layout id="ALL_3" filename="default_4x4_bar.xml">Bar 4x4</Layout>
    <Layout id="ALL_4" filename="default_4x4_chart.xml">Chart 4x4</Layout>
  </LayoutsForAllMachines>
  <MachineSpecificLayouts>
    <MachineSpecificLayout mnr="xyz">
      <Layouts>
      </Layouts>
    </MachineSpecificLayout>
  </MachineSpecificLayouts>
</Allocations>

```

XML element	Description
LayoutsForAllMachines	Provides a list of layouts that are available with all machines.
MachineSpecificLayouts	Includes the configurations of the layouts for the machines.
MachineSpecificLayout	Provides a list of layouts that are only available with the machines defined in "mnr".
Layout	Describes an individual layout: id: Internal ID for the layout assignment. Must be unambiguous. Filename: File name of the layout. Value: Display name of the layout. e.g. "Mix Standard Layout"

Customization:

- **In the MOC:**

You can store an individual layout assignment in the HYDRA server when using the MOC. It must be named "MachineLayoutConfiguration_custom.xml".

- **In the terminal:**

In the terminal, you can create specific layouts for a terminal or a terminal group. The storage of layout configurations is performed according to the default storage of configurations in the terminal. You can find the layouts in the terminal directory in **.etc\PDV\Layouts**.



The file contents are NOT merged.

Configuration of a layout

Sample configuration:

```
<LayoutGroup>
  <GroupName></GroupName>
  <ItemOrientation>Horizontal</ItemOrientation>
  <Items>
    <LayoutItem>
      <Type>Gauge</Type>
      <MaxValueCount>5</MaxValueCount>
      <Topic>mv/pp/%WORKPLACE%/%PPARAM%</Topic>
      <GuiRefreshTime>2000</GuiRefreshTime>
      <SmoothValueChange>true</SmoothValueChange>
      <MinorTickCount>10</MinorTickCount>
      <MajorTickCount>5</MajorTickCount>
      <Layout>CleanWhite</Layout>
    </LayoutItem>
  </Items>
</LayoutGroup>
```

General, valid for all types of display.

XML element	Description
Type	Specifies the type of display Linear: display of level indicator Digital: digital display Gauge: display of gauge chart Chart: display of trend line If you do not select a display type or the selected type is not found, a gauge chart is automatically

	displayed.
Topic	Specifies the MQTT topic. This specification is mandatory and is currently always "mv/pp/%WORKPLACE%/PPARAM%". The placeholders %WORKPLACE% and %PPARAM% are replaced with the respective PDV configuration during run time.
GuiRefreshTime	This property modifies the update interval [ms] of the control. Depending on the number of controls (i.e. the process parameters) and the frequency of incoming values, you can increase or decrease the update rate. Example: Many values + many process parameters = higher update interval Many values + few process parameters = medium update interval Few values + many process parameters = medium update interval Few values + few process parameters = low update interval Important: - The intervals also depend on the hardware equipment of the terminal or computer. - The controls store only the last value that was received. Exception: the chart display (trend line). If further values are received between the updates, the values are discarded and only the most recent value is adopted. The chart saves all incoming values.

Layout	Type of layout. See available layouts.
GroupName	Name of the layout group This name is also displayed in the view.
ItemOrientation	Horizontal: The controls are displayed in a row. Vertical: The controls are displayed in a column.

For charts:

XML element	Description
MaxValueCount	Identifies the maximum number of values to be displayed within the chart.

For all display types except chart:

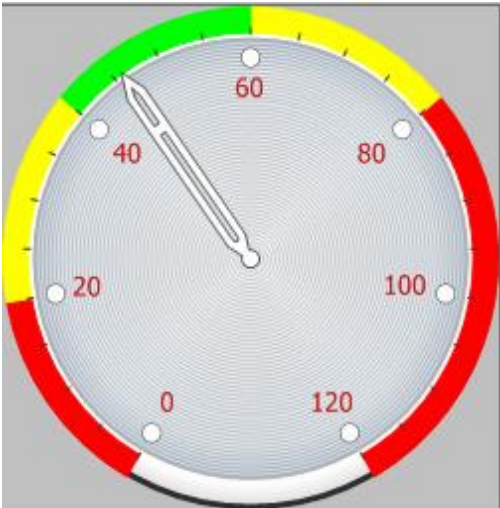
XML element	Description
SmoothValueChange	Enables a smooth transition of the pointer from one value to the next. Important: In case of high update rates, you should not use this feature as the smooth transition takes time.
MajorTickCount (not available with <i>Digital</i>)*	It is possible to overwrite the MajorTickCounts calculated during run time with a fixed value.
MinorTickCount (not available with <i>Digital</i>)*	This property enables a configuration of the number of ticks.

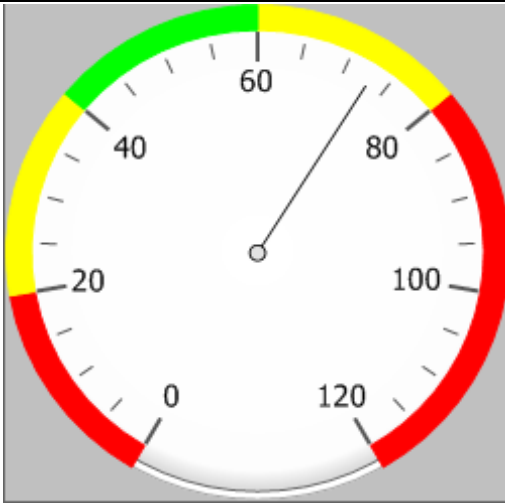
* A tick corresponds to a line in the scale. Therefore, a MajorTick corresponds to a line with a value. The MinorTicks are displayed between the MajorTicks.

Available layouts:

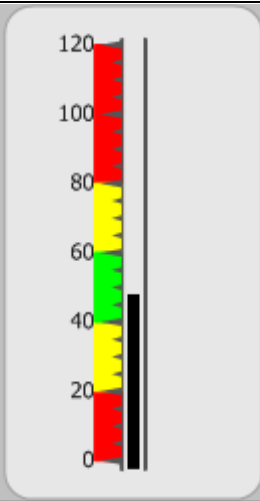
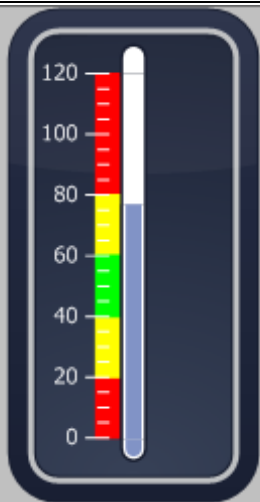
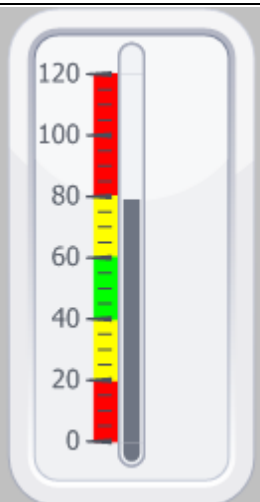
Layout	Display on AIP
--------	----------------

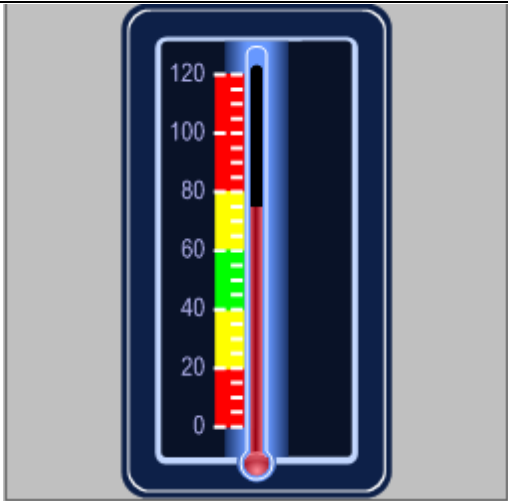
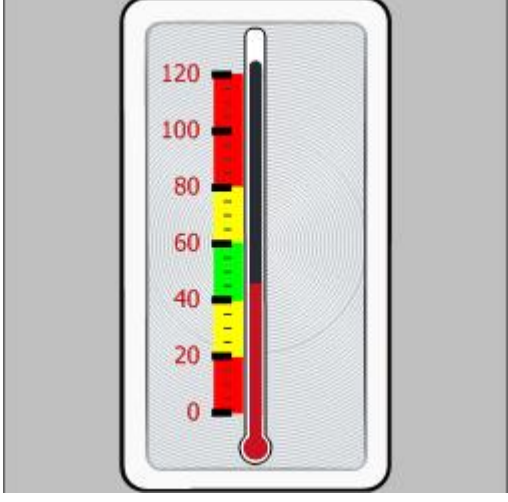
CleanWhite	 A speedometer with a white face and black markings. The needle is positioned at approximately 45. The background is divided into three colored segments: green (0-40), yellow (40-80), and red (80-120). The needle is black and the center is a small black circle.
Classic	 A speedometer with a dark blue face and white markings. The needle is positioned at approximately 75. The background is divided into three colored segments: green (0-40), yellow (40-80), and red (80-120). The needle is white and the center is a small white circle.
Clever	 A speedometer with a light gray face and black markings. The needle is positioned at approximately 75. The background is divided into three colored segments: green (0-40), yellow (40-80), and red (80-120). The needle is black and the center is a small black circle.

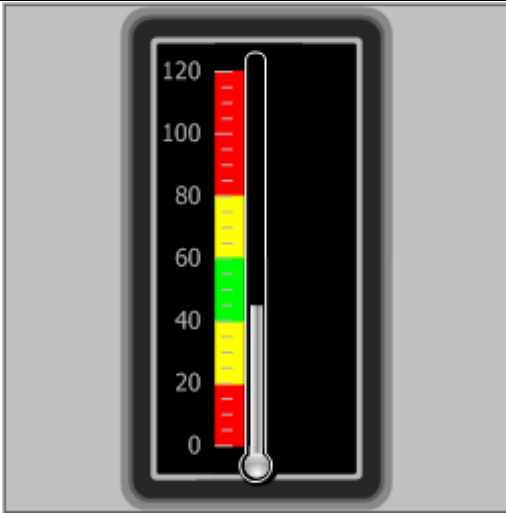
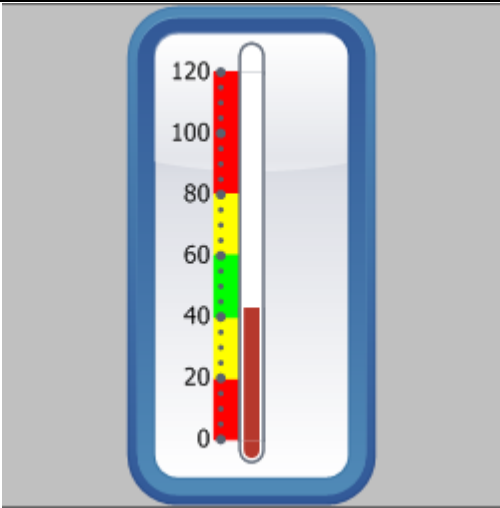
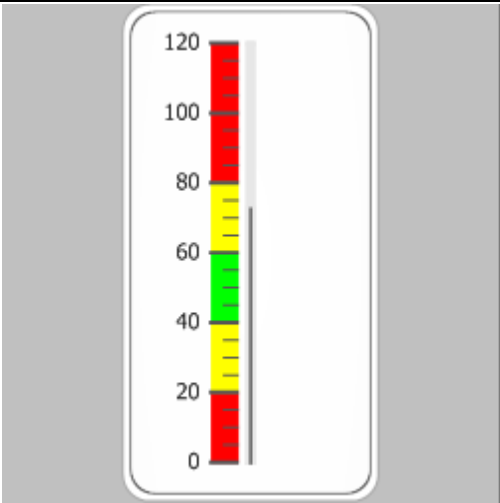
DarkNight		
iStyle		
Retro		

ShiningDark		
Smart		
White		

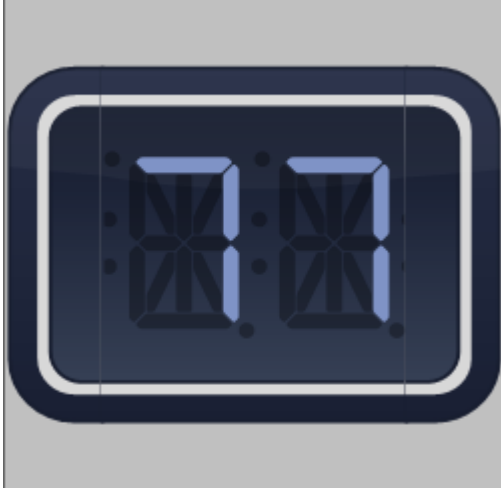
Layout	Display on AIP and MOC
--------	------------------------

CleanWhite		
Classic		
Clever		

DarkNight	 The DarkNight gauge features a dark blue background with a glowing blue border. The scale ranges from 0 to 120, with major tick marks every 20 units. The gauge is divided into four color-coded zones: red (0-40), yellow (40-60), green (60-80), and blue (80-120). The needle is positioned at approximately 75, indicating a value in the green zone.
iStyle	 The iStyle gauge has a light gray background with a black border. The scale ranges from 0 to 120, with major tick marks every 20 units. The gauge is divided into four color-coded zones: red (0-40), yellow (40-60), green (60-80), and blue (80-120). The needle is positioned at approximately 65, indicating a value in the green zone.
Retro	 The Retro gauge features a light gray background with a white border. The scale ranges from 0 to 120, with major tick marks every 20 units. The gauge is divided into four color-coded zones: red (0-40), yellow (40-60), green (60-80), and blue (80-120). The needle is positioned at approximately 45, indicating a value in the yellow zone.

ShiningDark		
Smart		
White		

Layout	Display on AIP and MOC
--------	------------------------

CleanWhite	
Classic	
Clever	

DarkNight	
iStyle	
Retro	

ShiningDark		
Smart		
White	